

1. Employees Above Their Department Average Display emp_id, emp_name, dept_id, and salary for employees whose salary is greater than the average salary of their own department. Do not use JOIN. Focus: correlated subquery with AVG().
2. Highest Paid Employee(s) in Every Department Display the highest-paid employee or employees from each department. If multiple employees in the same department share the highest salary, return all of them. Focus: correlated subquery with MAX().
3. Second Highest Distinct Salary Find the second-highest distinct salary in the company without using LIMIT, OFFSET, or JOIN. Focus: nested scalar subquery.
4. Employees Receiving the Second Highest Distinct Salary Display the employee name and salary of every employee whose salary equals the second-highest distinct salary in the company. Focus: nested subqueries and duplicate-safe result.
5. Employees Earning More Than Their Manager Display employees whose salary is greater than the salary of their own manager. Employees without a manager should not appear. Do not use a self JOIN. Focus: correlated scalar subquery against the same table.
6. Managers Who Earn Less Than At Least One Direct Report Display employees who are managers and whose salary is less than at least one employee reporting directly to them. Focus: correlated subquery with EXISTS or ANY.
7. Departments Whose Average Salary Is Above Company Average Display the department IDs whose average employee salary is greater than the overall average salary of all employees. Focus: aggregate query with a scalar subquery in HAVING.
8. Employees in Departments Whose Average Salary Is Above Company Average Display all employees who belong to departments whose departmental average salary is greater than the overall company average salary. Do not use JOIN. Focus: nested multiple-row subquery.
9. Employees Earning More Than Every Employee in HR Display employees whose salary is greater than the salary of every employee in the HR department. The department ID must be obtained using a subquery from the departments table. Focus: nested subquery with ALL.
10. Employees Earning More Than At Least One Research Employee Display employees whose salary is greater than at least one employee working in the Research department. Obtain the Research department ID through a subquery. Focus: nested subquery with ANY.
11. Departments Having No Projects Display dept_id and dept_name for departments that do not have any project listed in the projects table. Solve using a subquery only. Focus: NOT EXISTS correlated subquery.
12. Employees Belonging to Departments With More Than One Project Display all employees who belong to a department that has more than one project. Do not use JOIN. Focus: correlated / grouped subquery with COUNT().

13. Employees in the Department of the Highest-Budget Project Display all employees who work in the department that owns the project having the maximum budget. If multiple projects tie for maximum budget, employees from all matching departments should be returned. Focus: nested multi-row subquery with MAX().

14. Employee(s) With Salary Closest to Company Average Display the employee or employees whose salary has the smallest absolute difference from the company average salary. Do not use ORDER BY ... LIMIT and do not use JOIN. Focus: nested aggregate subqueries using AVG(), ABS(), and MIN().

15. Departments Whose Top Salary Exceeds Every Manager Salary in That Department Display department IDs where the maximum employee salary in that department is greater than the salary of every employee in the same department who is acting as a manager of at least one other employee. Focus: advanced correlated and nested subqueries with MAX(), ALL / NOT EXISTS.

Queries

-- q 1;

```
select e.emp_id,e.emp_name,e.dept_id,e.salary from employees e where e.salary > (select
avg(salary) from employees where dept_id=e.dept_id);
```

-- q 2

```
select e.emp_id,e.emp_name,e.dept_id,e.salary from employees e where salary = (select
max(salary) from employees where dept_id=e.dept_id);
```

-- q3

```
select max(salary) from employees where salary < (select max(salary) from employees);
```

-- q4

```
select emp_id,emp_name,salary from employees where salary =
(select max(salary) from employees where salary < (select max(salary) from employees));
```

-- q5

```
select emp_id,emp_name,salary from employees e where
```

```
salary > (select salary from employees where emp_id=e.manager_id);
```

```
-- q6
```

```
select emp_id,emp_name,salary from employees e where salary < any (select salary from  
employees  
where manager_id=e.emp_id);
```

```
-- q7
```

```
select distinct dept_id from employees group by dept_id having avg(salary) > (select  
avg(salary) from employees);
```

```
-- q8
```

```
select emp_id,emp_name,dept_id from employees where dept_id in  
(select dept_id from employees group by dept_id having avg(salary)> (select avg(salary)  
from employees));
```

```
-- q9
```

```
select emp_id, emp_name, salary from employees where salary >  
all(select salary from employees d where dept_id in (select dept_id from departments  
where dept_name='HR'));
```

```
-- q10
```

```
select emp_id, emp_name, salary from employees where salary >  
any(select salary from employees d where dept_id in (select dept_id from departments  
where dept_name='Research'));
```

```
-- q11
```

```
select dept_id, dept_name from departments d where dept_id in(select dept_id from  
projects where dept_id = d.dept_id);
```

-- q12

```
select emp_id,emp_name,dept_id from employees where dept_id in (select dept_id from projects group by dept_id having count(project_id)>1);
```

-- q13

```
select emp_id,emp_name from employees where dept_id in (select dept_id from projects where budget = (select max(budget) from projects));
```

-- q14

```
select emp_id,emp_name,salary from employees where abs(salary - (select avg(salary) from employees))= (select min(abs(salary-(select avg(salary) from employees))) from employees);
```

-- q15

```
SELECT dept_id FROM employees e GROUP BY dept_id HAVING MAX(salary) > ALL (SELECT m.salary FROM employees m WHERE m.dept_id = e.dept_id AND m.emp_id IN (SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL));
```